



BizOp: Gaming Company Cohort Analysis and Forecast

Overview

ABC Company ("The Company") is a leading platform for immersive gaming experiences, enabling users to create, share, and explore virtual worlds. The Company serves a diverse and active user base, with a core audience predominantly between the ages of **9 and 16**. The Company currently captures approximately **3% of the U.S. mobile and tablet gaming market**.

Objective

This project is divided into two key parts:

Part I focuses on analyzing historical trends in user growth and revenue for The Company from 2016 to 2021 ("The Observation Period"). The goal is to derive actionable insights and develop theories explaining key findings based on the provided data.

Part II leverages the insights from Part I, combined with market research, to create a detailed forecast for monthly active users (“MAU”) and revenue for 2022.

Data Provided

Two sets of data was provided. **Dataset I** (on the left) contains user information including LoginMonth, SignUpMonth, SignUpYear, and MAUCount. **Dataset II** (on the right) contains billing information including PaymentMonth, SignUpMonth, MonthlyUniquePayer (“MUP”) Count, and Billing.

1	LoginMonth	SignUpMonth	SignUpYear	MAUCount
2	1/1/2022	4/1/2020	2020	13,587
3	1/1/2022	1/1/2021	2021	26,595
4	1/1/2022	8/1/2021	2021	58,935
5	1/1/2022	5/1/2021	2021	33,737
6	1/1/2022	12/1/2020	2020	24,351

MAU by Month		Billing by Month	
Date	MAU	Date	Billings
Jan-16	1,721,373	Jan-16	1,498,926
Feb-16	1,759,401	Feb-16	1,327,293
Mar-16	1,981,070	Mar-16	1,422,477
Apr-16	2,043,796	Apr-16	1,525,200

	A	B	D	E
1	PaymentMonth	SignUpMonth	MUPCount	Billings
2	1/1/2022	5/1/2021	2,555	\$50,263.20
3	1/1/2022	4/1/2020	1,180	\$20,910.23
4	1/1/2022	8/1/2021	4,174	\$81,652.43
5	1/1/2022	1/1/2021	2,260	\$40,338.27

The datasets were provided in .xlsx. format and were not chronologically sorted. Not to mention that the datasets included a lot of extra information that lies outside of the scope of research.

Step 1: Data Cleaning

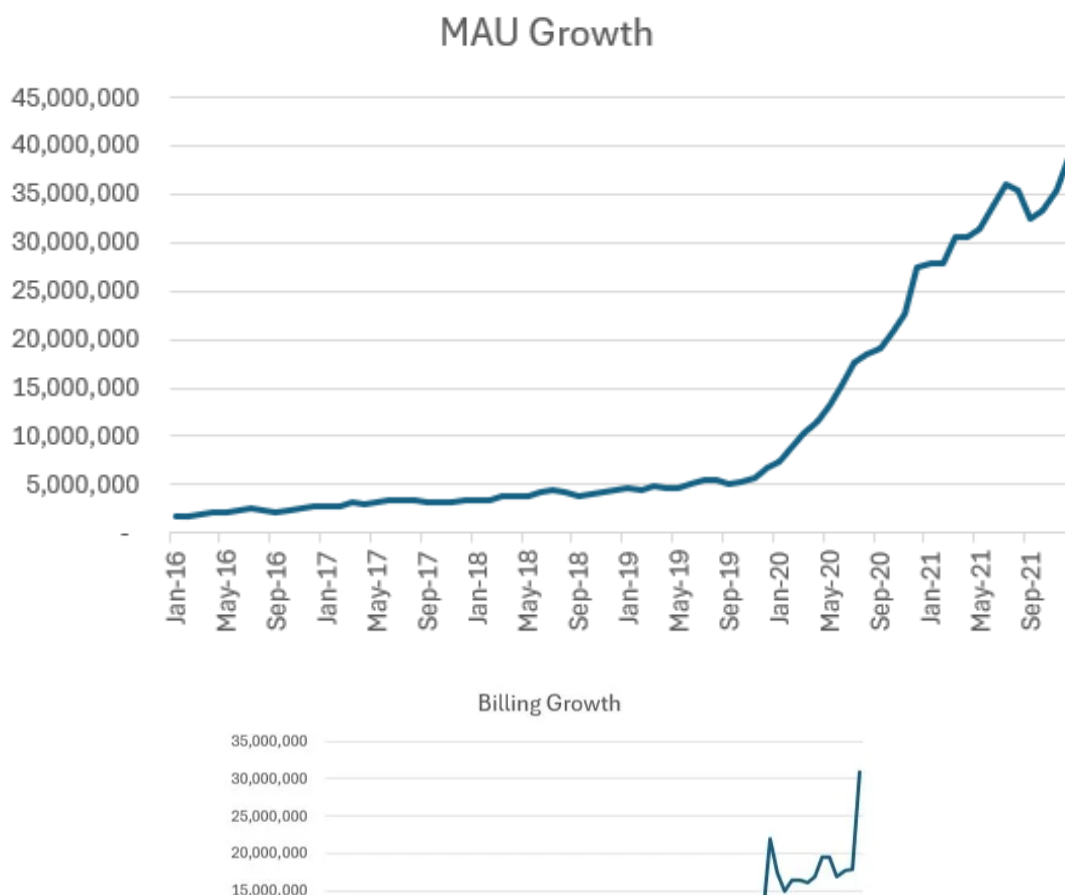
In order to get a better understanding of the data provided, I immediately dove into cleaning the data and getting rid of junk information that lies outside of the scope of my analysis.

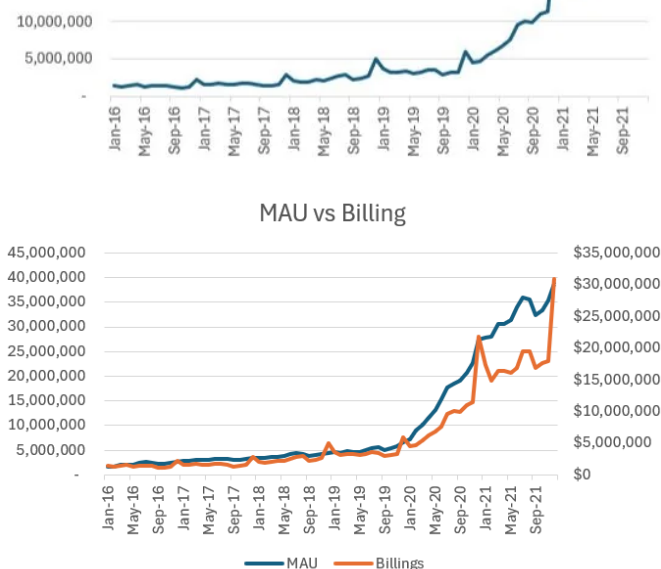
Since the Company is only interested in its user and billing performance over the 2016 to 2021 period, I utilized applicable functions such as **SUMIF** to extract the total number of MAU and billing on a month-to-month basis. This gives me a clear understanding of the **total number of users that logged-in** and the **total amount of billing revenue accrued** in a specific month.

Organizing the datasets in this **chronological order** also provides me with better visual clarity on the **growth trends** of the parameters under interest.

Step 2: Analyzing the Historical Trend

Next, I mapped month-to-month data and discovered some interesting things occurred over the Observation Period.





First, looking at the line chart analysis of the MAU growth reveals a period of stagnation from 2016 to 2019, followed by a **significant surge starting in 2020**. This sharp increase aligns with the onset of the **COVID-19** pandemic, as limited indoor activities drove an increase in screen time.

By the end of 2021, the platform had surpassed **27 million** MAUs.

For billing growth, a drastic increase also began in early 2020, with the platform now surpassing **\$21 million** in billing revenue by the end of 2021.

Interestingly, the line chart reveals **spikes in December of each year**, highlighting a **seasonality trend** in billing and monetization.

I hypothesize that the December surge in billing revenue is influenced by an **increase in disposable income** during the holiday season. For young users under 21, this may be driven by the redemption of gift cards received as holiday presents. For adult users over 21, year-end bonuses likely contribute to higher in-game purchases during this period.

To better understand the relationship between MAU growth and billing growth, I stacked the two line charts to observe their cohesion.

From this visualization, I noted a **widening gap between MAU growth and billing growth through 2021**, suggesting that user count was accelerating faster than monetization. This disparity might indicate a strategic focus on user acquisition while monetization efforts lagged, potentially due to a lack of emphasis on creating valuable in-game items that incentivize user purchases.

However, by the end of 2021, billing experienced a **vertical acceleration** and closed the gap with MAU growth. This shift highlights a **positive value creation** for

players, reflecting the platform’s success in enhancing monetization strategies.

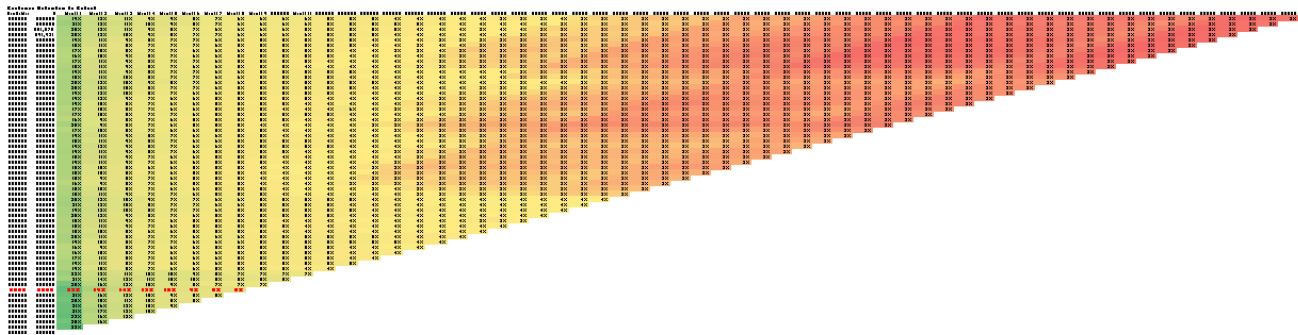
Step 3: Understanding Retentions

Next, I want to better understand user retention. For a gaming company, it is crucial to examine user retention by **segmenting users based on their onboarding date**, also known as a **Cohort Analysis**. Cohort Analysis is critical to understanding player behavior and lifecycle. It helps identify trends in user retention, gauge the effectiveness of updates or promotions, and evaluate how different player segments interact with the game. This insight enables companies to optimize user acquisition strategies, improve in-game features, and boost monetization, ensuring a better overall gaming experience and sustained revenue growth.

I conducted **two seperate** Cohort Analyses to assess the MAU retention and MUP retention independently.

Below is a heatmap-style cohort analysis visually tracks **MAU retention** over time:

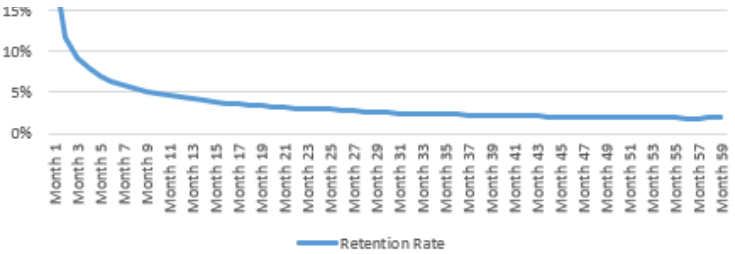
Gaming Company Cohort Analysis and Forecast



Axes:

- The **horizontal axis** represents time in months (0–59) since onboarding (Month 0 being the start).
- The **vertical axis** represents cohorts grouped by onboarding date, with each row corresponding to users who started in a specific month.





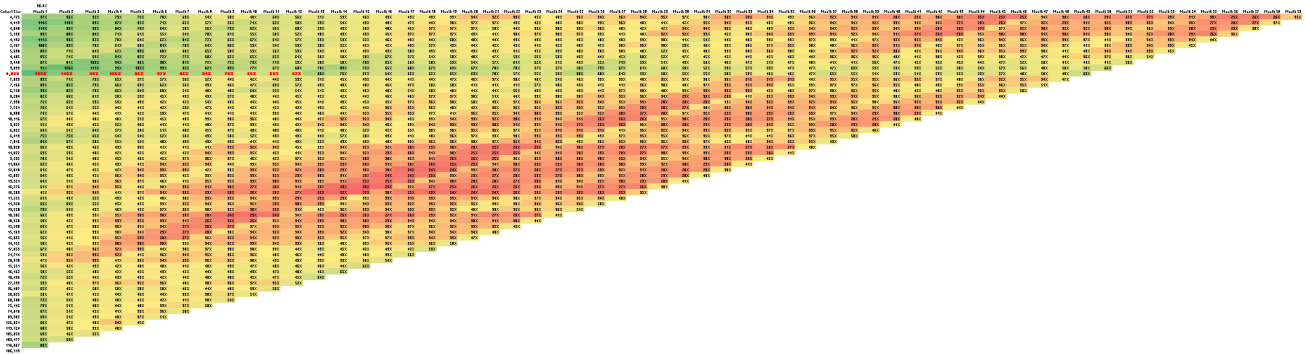
Colors:

- The heatmap uses a color gradient (green to red) to represent **retention rates**, with **green indicating higher retention** and **red indicating lower retention**.

If we look at the average retention rate across cohorts from Month 0 to Month 59 after onboarding, we quickly see that most cohorts experience a **sharp decline** in retention **within the first few months**. This is typical in many user-based platforms where only a small fraction of users remain engaged long-term.

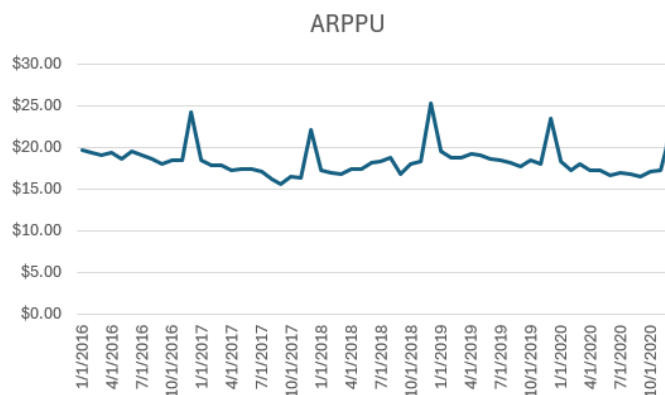
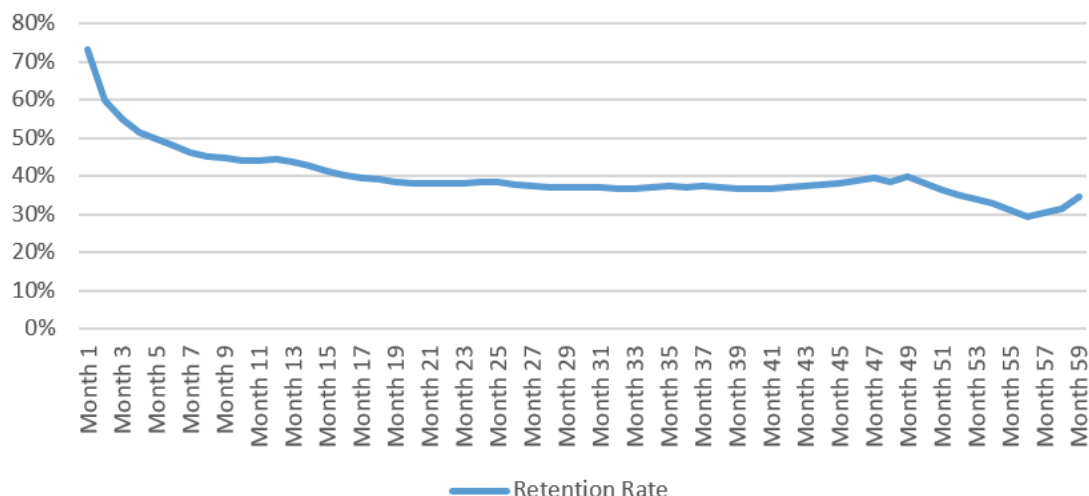
After Month 12 of joining the platform, MAU retention rate **stabilizes** between the range of 1.0% to 5.0%. These are the **loyal players** who stayed with the platform.

Similarly, for **MUP retention**, I leveraged a heatmap-style cohort analysis to understand the retention rate of paying users **after initial payment** on the platform.



For paying users, retention exhibits a distinct pattern compared to overall users. While there is a slight decline in the first few months after their initial payment, paying users demonstrate **higher stickiness and loyalty**. After 12 months, their retention rate stabilizes between 35% and 45%, indicating that many paying users find the platform worthwhile and continue to invest in it.

However, an interesting **drop** in retention occurs around **Month 49** after the initial payment. I hypothesize that this decline may stem from a **game design issue**.



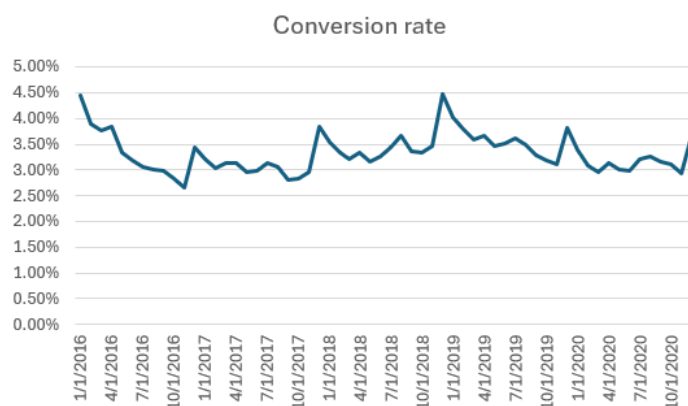
A **game design issue** can significantly impact the retention of paying users, causing them to disengage and stop spending. For instance, a **progression barrier**—where advancing in the game becomes disproportionately difficult or requires excessive monetary investment—can create frustration among users. Additionally, a **content gap**, such as a lack of new features, levels, or engaging updates, can lead to a repetitive and uninspiring gaming experience. Paying users, who invest money into the platform, often have higher expectations for sustained value and engagement. When the game fails to meet these expectations, they may perceive it as no longer worthwhile, leading to reduced retention and spending. This decline is notably evident in the drop in retention around **Month 49** after initial payment, which may be attributed to such design flaws.

To address these challenges, it is crucial to implement a series of strategic improvements. **Regular content updates**, including the introduction of new levels, features, and limited-time events, can maintain excitement and keep the experience fresh. A **progression system review** is also necessary to ensure that the game's difficulty curve and rewards are balanced, making progression achievable and rewarding without requiring excessive effort or spending.

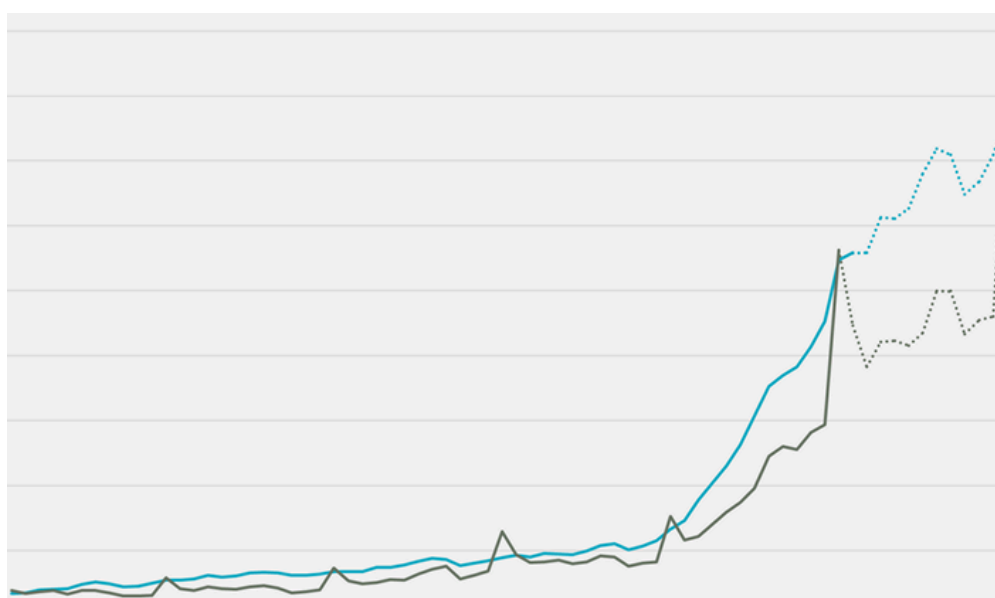
Additionally, actively soliciting **player feedback** can help identify specific frustrations and provide insights for tailoring updates to meet user needs. Finally, offering **targeted promotions**—such as discounts, exclusive items, or bonus rewards—can re-engage paying users and incentivize them to continue investing in the platform.

Step 4: Understanding Monetization KPIs

One of the most important monetization KPI to analyze is the **non-paying to paying user conversion rate**, which is crucial for a gaming company as it directly drives revenue and reflects the effectiveness of monetization strategies.



Applying these growth rates to our ending MAU and billing, I arrive at the **2022 forecast** for the Company.



The Company demonstrates an **average conversion rate of 3.3%** over the Observation Period, with fluctuations ranging from a minimum of 2.7% to a

maximum of 4.5%. While the overall trend reflects relatively stable conversion rates, occasional spikes likely correspond to promotions, new features, or seasonal activity.

However, it is notable that there are **no significant breakthroughs in the conversion rate**, which may indicate a **lack of major value incentives** compelling users to transition from non-paying to paying. This stagnation suggests an opportunity for the Company to introduce more innovative or enticing features to drive higher conversion rates and unlock additional revenue potential.

Another important monetization KPI is the **Average Revenue Per Paying User ("ARPPU")**. ARPPU is a critical metric as it reflects the revenue generated per paying customer, helping to gauge the effectiveness of monetization strategies. It provides insight into how much paying users are willing to spend and helps identify monetization opportunities.

The Company's **ARPPU** over the Observation Period is **\$18.41**, with noticeable **spikes in December** each year. These seasonal peaks are consistent with the hypothesis that holiday promotions, discounts, and seasonal in-game events drive heightened activity and spending.

Using the ARPPU of \$18.41 and the average retention rate of a paying user (38.9%), the **Lifetime Value (LTV) of a paying user** can be calculated with the formula: **$LTV = ARPPU / (1 - retention\ rate)$** , resulting in an LTV of **\$30.14**.

Taking it a step further, the **LTV of all users** can be determined by **multiplying the LTV per paying user by the non-paying to paying user conversion rate (3.3%)**, which gives an overall user LTV of **\$0.99**.

This figure is particularly important because it represents the **maximum amount the company should spend on user acquisition** to ensure profitability, which, in this case, is \$0.99 per user. Understanding this metric is critical for optimizing marketing budgets, designing effective user acquisition strategies, and sustaining long-term financial growth.

Step 5: Future Forecast

In order to create the 2022 MAU and billing forecast, I first looked at the historical month-to-month average growth rate in each category.

For **MAU** historical growth rate, I **excluded** the exponential growth data experienced in 2020, in order to get an unbiased understanding of the growth pattern from month-to-month.

MAU Forecast	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
2016			2.2%	12.6%	3.2%	3.1%	13.9%	7.6%	-5.1%	-9.1%	1.7%	10.1%	9.8%
2017	0.8%	1.8%	10.0%	-3.3%	3.2%	7.3%	0.8%	-0.4%	-6.5%	-0.1%	2.9%	6.2%	
2018	0.6%	-0.4%	9.3%	0.7%	4.7%	7.0%	5.8%	-2.6%	-10.2%	4.4%	4.4%	5.5%	
2019	4.6%	-3.5%	6.7%	-1.4%	-0.6%	5.8%	8.5%	2.6%	-8.4%	6.0%	7.1%	15.6%	
Average	2.0%	0.0%	9.7%	-0.2%	2.6%	8.5%	5.7%	-1.4%	-8.6%	3.0%	6.1%	9.3%	

For **billing** historical growth rate, I just looked at the historical average month-to-month growth rate over the entire Observation Period. It is important to analyze on a month-to-month basis since seasonality is present.

Billing forecast	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
2016			2.2%	12.6%	3.2%	3.1%	13.9%	7.6%	-5.1%	-9.1%	1.7%	10.1%	9.8%
2017	0.8%	1.8%	10.0%	-3.3%	3.2%	7.3%	0.8%	-0.4%	-6.5%	-0.1%	2.9%	6.2%	
2018	0.6%	-0.4%	9.3%	0.7%	4.7%	7.0%	5.8%	-2.6%	-10.2%	4.4%	4.4%	5.5%	
2019	4.6%	-3.5%	6.7%	-1.4%	-0.6%	5.8%	8.5%	2.6%	-8.4%	6.0%	7.1%	15.6%	
2020	9.8%	21.9%	14.7%	12.7%	14.2%	16.6%	15.1%	4.7%	3.6%	8.1%	9.4%	21.0%	
Average	3.9%	4.4%	10.7%	2.4%	4.9%	10.1%	7.6%	-0.2%	-6.1%	4.0%	6.8%	11.6%	

The 2022 forecast projects The Company to reach **38.7 million Monthly Active Users (MAUs)**, **1.42 million Monthly Unique Payers (MUPs)**, and generate **\$31 million in billings**. The forecast shows a steady increase in MAUs and billings, with an accelerated growth in billing in December 2026 due to a consistent seasonality trend.

Summary

The analysis of The Company's performance reveals several key insights into its growth, retention, and monetization strategies. The platform demonstrated a steady rise in Monthly Active Users (MAUs) from 2020, driven by factors such as the pandemic's impact on screen time, reaching 27 million MAUs by the end of 2021. However, billing growth lagged behind MAU growth until late 2021, suggesting a focus on user acquisition over monetization. This gap was closed by a surge in billing, reflecting improved value creation for users.

The MAU to MUP conversion rate averaged 3.3% over the observation period, with spikes during December due to holiday promotions and seasonal in-game

events. However, the absence of significant breakthroughs in conversion rates points to untapped potential in driving non-paying users to become paying customers. Meanwhile, the Average Revenue Per Paying User (ARPPU) was \$18.35, also exhibiting seasonal spikes, reinforcing the impact of holiday-related activities.

The calculated Lifetime Value (LTV) of a paying user is \$30.14, and when adjusted for the non-paying to paying conversion rate, the average LTV across all users is \$0.99. This figure represents the maximum sustainable cost for user acquisition, emphasizing the need for efficient marketing strategies.

Retention analysis revealed that paying users showed greater loyalty, with retention rates stabilizing between 35% and 45% after 12 months. However, a notable drop around Month 49 suggests a potential game design issue, such as progression barriers or lack of engaging updates.

2022 forecasts predict further growth, with 38.7 million MAUs, 1.42 million MUPs, and \$31 million in billings, indicating strong prospects for user acquisition and monetization if current trends continue.

Suggestions for Improvement:

1. **Enhance Conversion Rates:** Develop compelling value incentives to convert more non-paying users, such as exclusive in-game features or limited-time promotions.
2. **Content Updates:** Regularly introduce new levels, features, and seasonal events to sustain user engagement and prevent stagnation.
3. **Retention Optimization:** Address game design issues by refining progression systems and incorporating user feedback to eliminate potential barriers.
4. **Data-Driven Promotions:** Leverage insights from seasonal spikes to create targeted campaigns during off-peak months.
5. **User Acquisition Strategy:** Align marketing spend with the \$0.99 LTV of all users to ensure sustainable growth.

By addressing these areas, The Company can further capitalize on its growing user base, enhance monetization, and strengthen retention, positioning itself for long-term success.

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